

Code Structure - Simple Hinglish Summary

Source: <https://javascript.info/structure>

Main baat

Sabse pehle code ke **building blocks** seekhte hain: **Statements, Semicolons aur Comments.**

1. Statements

- Statement wo command hai jo **koi action perform karta hai**. Jaise `alert('Hello, world!')`.
- Jitne chahe statements likh sakte ho. Unhe **semicolon (;)** se alag karte hain.

```
javascript  
  
alert('Hello'); alert('World');
```

- Readability ke liye **har statement alag line me** likhte hain:

```
javascript  
  
alert('Hello');  
alert('World');
```

2. Semicolons

- Zyadatar cases me **line break ho to semicolon likhna zaruri nahi**. JavaScript khud semicolon laga leta hai. Ise **Automatic Semicolon Insertion (ASI)** kehte hain.

```
javascript  
  
alert('Hello')  
alert('World')
```

- **Lekin "zyadatar" ka matlab "hamesha" nahi hota.**
- Kuch cases me newline ka matlab semicolon nahi hota. Jaise jab line **(+)** par khatam ho (incomplete expression):

```
javascript
```

```
alert(3 +  
1  
+ 2); // output: 6
```

Error ka example (semicolon miss hone par)

Ye sahi chalta hai ((Hello), (1), (2) dikhata hai):

```
javascript  
  
alert("Hello");  
  
[1, 2].forEach(alert);
```

Semicolon hata diya to **error** aata hai:

```
javascript  
  
alert("Hello")  
  
[1, 2].forEach(alert);
```

Kyu? JavaScript **square brackets** ([...]) se pehle semicolon nahi lagata, isliye wo isko ek hi statement samajhta hai:

```
javascript  
  
alert("Hello")[1, 2].forEach(alert);
```

Aise errors dhundhna **bahut mushkil** hota hai.

Recommendation

Semicolon hamesha lagao, chahe newline ho. Ye community ka common rule hai aur beginners ke liye safe hai.

3. Comments

Program bade hote hain, to code ke saath **explanation (kya aur kyu)** likhna zaruri ho jata hai. Comments ko JS engine **ignore** karta hai, execution par koi effect nahi.

Single-line comment: (//)

```
javascript
```

```
// Ye poori line comment hai
alert('Hello');

alert('World'); // Ye statement ke baad wala comment hai
```

Multi-line comment: `/* ... */`

```
javascript

/* Do messages ka example.
Ye multiline comment hai.
*/
alert('Hello');
alert('World');
```

Code ko temporarily band karna

```
javascript

/* Commenting out the code
alert('Hello');
*/
alert('World'); // sirf ye chalega
```

Hotkeys (bahut kaam ke)

Kaam	Windows/Linux	Mac
Single-line comment	<code>Ctrl + /</code>	<code>Cmd + /</code>
Multi-line comment	<code>Ctrl + Shift + /</code>	<code>Cmd + Option + /</code>

Nested comments allowed nahi

`/* ... */` ke andar dusra `/* ... */` nahi likh sakte, error aayega.

```
javascript

/*
  /* nested comment !? */
*/
alert( 'World' ); // ERROR
```

Comments likhne se darna nahi

Comments code ka size badhate hain, lekin problem nahi. Production me **minify tools** comments hata dete hain, isliye koi nuksan nahi.

Quick Summary

Topic	Yaad rakhne wali baat
Statement	Ek action wali command, <code>(;)</code> se alag hoti hai
Semicolon	Optional lagta hai, lekin hamesha lagao (safe)
<code>//</code>	Single-line comment
<code>/* */</code>	Multi-line comment (nested nahi chalta)
Comments	Engine ignore karta hai, likhne me hichkichao mat